

RDCC and Euclid: A Survey-Ready Framework for Testing Relaxation-Driven Cosmology

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Abstract

The Relaxation-Driven Cyclic Cosmology (RDCC) introduces a scale-dependent relaxation parameter α that modifies the gravitational potential and induces distinct geometric and topological signatures in weak-lensing observables. This paper provides a Euclid-focused overview of RDCC, summarizing its theoretical foundations, its observational signatures, and its compatibility with Euclid’s survey design, data products, and analysis pipelines. We present Euclid-level forecasts, identify optimal RDCC probes, and outline a practical integration strategy for testing RDCC within the Euclid weak-lensing and large-scale structure programs. This document is intended as a concise, survey-oriented reference for Euclid collaborators and theorists.

1 Introduction

The Euclid mission provides an unprecedented opportunity to test extensions of Λ CDM through high-precision weak-lensing and galaxy-clustering measurements. The Relaxation-Driven Cyclic Cosmology (RDCC) is a theoretically motivated alternative that modifies the gravitational potential via a scale-dependent relaxation parameter α . This modification induces geometric, morphological, and topological signatures in the convergence field that are directly accessible to Euclid’s weak-lensing pipeline.

This paper summarizes the RDCC framework from a Euclid perspective. We focus on: (i) the theoretical structure of RDCC, (ii) its observational signatures in Euclid data products, (iii) Euclid-level forecasts for α , and (iv) practical integration pathways for testing RDCC within Euclid’s analysis infrastructure.

Nomenclature

α	RDCC relaxation parameter; controls scale-dependent modifications of the potential
Φ	Gravitational potential
Φ_{RDCC}	Modified potential in the RDCC model
$\Phi_{\Lambda\text{CDM}}$	Standard potential in the ΛCDM model
$\mathcal{R}(k, z; \alpha)$	RDCC relaxation kernel; scale- and redshift-dependent modification
k	Wavenumber in Fourier space
z	Redshift
κ	Weak-lensing convergence field
γ	Shear field
$P_\kappa(l)$	Convergence power spectrum as a function of multipole l
MF_i	Minkowski functionals (area, boundary length, Euler characteristic)
$\mathcal{D}$	Persistence diagram in persistent homology
W_2	Quadratic optimal-transport distance between persistence diagrams
$\sigma(\alpha)$	Forecasted uncertainty on the RDCC parameter α
$\Delta \log Z$	Evidence contrast between RDCC and ΛCDM
v_k	Fisher channel (eigenvector of the information-tensor decomposition)
λ_k	Eigenvalue of the k -th Fisher channel (information contribution)

2 RDCC: Theoretical Overview

RDCC modifies the gravitational potential through a relaxation kernel

$$\Phi_{\text{RDCC}}(k, z) = \Phi_{\Lambda\text{CDM}}(k, z) \cdot \mathcal{R}(k, z; \alpha),$$

where $\mathcal{R}$ introduces scale-dependent deviations that become relevant at $k \gtrsim 0.1 h \text{ Mpc}^{-1}$. These deviations propagate into the lensing potential, affecting:

- the convergence field κ ,
- shear correlations,
- peak and void statistics,
- Minkowski functionals,
- persistent homology and transport-based topology.

RDCC is fully compatible with Euclid’s tomographic binning and angular resolution, and its signatures are strongest at intermediate multipoles accessible to Euclid’s VIS instrument.

3 RDCC Signatures in Euclid Observables

RDCC induces characteristic distortions in Euclid weak-lensing data:

4.1 Power spectra

Moderate-scale suppression or enhancement depending on α ; partially degenerate with ΛCDM parameters.

4.2 Peak and void statistics

RDCC shifts peak heights and void depths, providing strong sensitivity to relaxation-driven potential changes.

4.3 Minkowski functionals

Morphological distortions in area, boundary length, and Euler characteristic.

4.4 Persistent homology

RDCC modifies the birth–death structure of topological features, producing robust, noise-resistant signatures.

4.5 Transport-based topology

Optimal-transport distances between persistence diagrams provide a high-sensitivity probe of RDCC-induced non-Gaussian structure.

4 Euclid-Level Forecasts

Using Euclid-like noise, resolution, and tomographic binning, RDCC forecasts yield:

$$\sigma(\alpha) \sim \mathcal{O}(10^{-2}),$$

with strongest constraints from:

- transport-based topological statistics,
- peak and void counts,
- neural-compressed summaries,
- combined multi-probe likelihoods.

Fisher-channel analysis shows that $\sim 70\%$ of RDCC information resides in the leading channel, enabling efficient compression and robust inference.

5 Degeneracy Structure and Euclid Synergy

RDCC variations can partially mimic Λ CDM parameter shifts. Euclid’s multi-probe design helps break these degeneracies:

- tomographic binning separates redshift-dependent relaxation effects,
- cross-correlation with galaxy clustering reduces null directions,
- topological probes provide strong orthogonality to Λ CDM-aligned manifolds.

6 Integration into Euclid Pipelines

RDCC can be tested within Euclid’s existing analysis framework:

- **Forward modeling:** RDCC can be incorporated into lensing simulations via modified potential kernels.
- **Summary statistics:** RDCC signatures are compatible with Euclid’s power-spectrum pipeline and can be extended to include peaks, Minkowski functionals, and topology.
- **Likelihood integration:** RDCC likelihood components can be added modularly to Euclid’s cosmological parameter estimation framework.
- **Validation:** The RDCC Benchmark Suite provides simulation-based calibration and cross-probe consistency tests.

7 Conclusion

RDCC provides a theoretically motivated and observationally testable extension of Λ CDM. Its geometric and topological signatures are well matched to Euclid’s weak-lensing capabilities, and Euclid-level forecasts demonstrate that RDCC can be constrained with high precision. This paper outlines a practical pathway for integrating RDCC into Euclid’s analysis pipelines and provides a survey-ready framework for testing relaxation-driven cosmology.